

Test Plan & Evidence

B.Tech CSE 2025–29 · Software Engineering & Project Management · Semester III

Dashboard basis: The project author has developed three dashboards—Student, Faculty and Admin. This document uses the supplied walkthrough to describe them and distinguishes those dashboard details from requirements, integrations or metrics that are proposed or not independently verified.

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1. Purpose and test scope

This plan tests the rules that most directly affect access, payment status, course progress and certificate eligibility. Test data should use non-production accounts and a payment-provider test environment. The supplied dashboard summary reports **18/18 automated tests passed, 16 internal defects caught, 1 defect found by a user, DRE 94.12%, and defect density 1.2 defects/KLOC**. The underlying test log, defect IDs and code-size measurement were not attached, so those figures are recorded as **reported evidence** rather than independently reproduced results.

2. Boundary-value analysis

Quiz score: 10 questions, pass at 70%

For 10 questions, the allowed integer number correct is 0–10; the threshold is 7 correct.

Test ID	Correct / 10	Expected score	Expected result	Purpose
BVA-Q1	-1 (invalid input)	—	Reject request; do not issue certificate	Below minimum / malformed boundary
BVA-Q2	0	0%	Fail	Minimum valid value
BVA-Q3	6	60%	Fail	Just below pass threshold
BVA-Q4	7	70%	Pass	Exact pass boundary
BVA-Q5	8	80%	Pass	Just above threshold

Test ID	Correct / 10	Expected score	Expected result	Purpose
BVA-Q6	10	100%	Pass	Maximum valid value
BVA-Q7	11 (invalid input)	—	Reject request; do not issue certificate	Above maximum / malformed boundary

Completion percentage

Test ID	Progress input	Expected quiz access	Purpose
BVA-P1	-1% (invalid)	Reject or clamp according to documented API validation; never unlock	Below minimum
BVA-P2	0%	Locked	Minimum valid value
BVA-P3	99%	Locked	Just below threshold
BVA-P4	100%	Unlocked only for an active, paid enrolment	Exact threshold / maximum
BVA-P5	101% (invalid)	Reject or clamp; never store >100	Above maximum

3. Equivalence classes

Field / rule	Equivalence class	Representative test	Expected behaviour
Quiz answers	0–6 correct	6 correct	Fail; no certificate
Quiz answers	7–10 correct	7 correct	Pass; proceed to certificate eligibility check
Quiz request	Missing, malformed, too few/many answers	9 answers	Reject as invalid; do not score or issue certificate
Progress	0–99%	50%, 99%	Quiz remains locked
Progress	Exactly 100%	100%	Quiz may be taken if enrolment is active and paid
Progress	Outside 0–100%	-1%, 101%	Reject or safely normalize; never grant eligibility from invalid input
Enrolment	Active and payment verified	Paid active learner	Learner can access lessons
Enrolment	Pending/failed/unpaid/cancelled	Pending payment	No protected lessons or certificate eligibility
User role	Student / assigned faculty / admin	One account per role	Permit only role-appropriate access

4. Certificate eligibility decision table

Condition / rule	R1	R2	R3	R4	R5
Enrolment active and payment verified?	Y	Y	Y	N	Y
Course progress = 100%?	Y	Y	N	Y	Y
Quiz submitted with valid 10 answers?	Y	Y	—	Y	Y
Score $\geq$ 70%?	Y	N	—	Y	Y
Issue certificate?	Yes	No	No	No	Yes
Expected result	Pass and one certificate	Fail; show score only	Quiz locked / reject submission	No protected eligibility	Pass; duplicate request returns same certificate

— means later conditions are not evaluated because the learner has not reached the quiz stage. Include a negative test that forges a client request for a certificate; the server must recompute eligibility from persisted enrolment/progress/attempt data.

5. Test cases and traceability

ID	Test / setup	Expected result	Related requirement
TC-01	Load catalogue with published courses	Course fields and ₹4,500 price display consistently	FR-01
TC-02	Register new valid learner and sign in	Account created; session assigned student role	FR-02
TC-03	Sign in with wrong password	Request rejected; no session created	FR-02, NFR-05
TC-04	Request enrolment for an available course	Pending record created with correct learner/course IDs	FR-03
TC-05	Send verified test payment response	₹4,500 transaction reference recorded; enrolment activated once	FR-04
TC-06	Open video as paid learner, then as another/unpaid user	Learner can stream; unauthorized user is denied	FR-05, NFR-01, NFR-03, NFR-04
TC-07	Complete lessons and refresh dashboard	Saved progress matches completed lessons	FR-06
TC-08	Attempt quiz at 99% progress	Quiz remains locked	FR-07
TC-09	Attempt quiz at 100% with active paid enrolment	Quiz becomes available	FR-07
TC-10	Submit 10 answers with known correct count	Score equals $\text{round}(\text{correct} \div 10 \times 100)$	FR-08

ID	Test / setup	Expected result	Related requirement
TC-11	Submit 6/10	60%, not passed; no certificate	FR-09, FR-10
TC-12	Submit 7/10	70%, passed; eligible for certificate	FR-09, FR-10
TC-13	Submit eligible result twice / forge request for ineligible learner	One certificate only for eligible learner; forged/ineligible requests denied	FR-10
TC-14	Faculty edits assigned and unassigned course question banks	Assigned update allowed; unassigned update denied	FR-11, NFR-04
TC-15	Compare admin metrics to source records	Counts reconcile; no role leakage	FR-12, NFR-04

6. Reported quality figures and calculation

Defect Removal Efficiency (DRE)

Reported figures: 16 defects caught internally before release and 1 defect caught by a user after release.

$DRE = \text{internal defects caught} \div (\text{internal defects caught} + \text{user-caught defects}) = 16 \div (16 + 1) = 16 \div 17 = 0.941176... = 94.12\%$

Defect density

The supplied dashboard states **1.2 defects/KLOC**. The standard calculation is:

$\text{Defect density} = \text{total counted defects} \div \text{delivered source size in KLOC}$

The exact source size and counted-defect population were not supplied. If (and only if) the 17 defects above are the full numerator, 1.2 defects/KLOC implies about $17 \div 1.2 = 14.17$ KLOC. This is an inference, not verified measurement. Attach the source-code size report and defect log to make the figure reproducible.

Test execution summary

The dashboard walkthrough reports **18/18 tests passed** and mentions boundary coverage at quiz scores 60%, 70%, 80% and progress 0%, 99%, 100%. This document expands the coverage into a proposed test suite. Until the actual execution log is attached, do not present the additional cases in the tables as already executed.

7. Defect and issue log

ID	Status	Description	Source / evidence	Follow-up
D-01 (details pending)	Reported escaped defect; closed status not evidenced	One user-caught defect is included in the supplied DRE numerator; its description, severity and fix are not provided	Dashboard summary only	Replace placeholder with actual issue ID, reproduction steps, severity, fix version and retest evidence
D-02 to D-17	Aggregate only	Sixteen internal defects are reported as caught before release; individual records not supplied	Dashboard summary only	Attach defect log to substantiate count and removal stage

8. Entry, exit and evidence rules

- **Entry:** stable test build, test accounts, seeded course and quiz data, provider sandbox and accessible test environment.
- **Exit:** all Must requirements pass; no open critical/high defect; certificate eligibility and role-isolation negatives pass; failed tests are logged with reproducible steps.
- Preserve build identifier, test date, tester, request/response evidence with secrets removed, defect IDs and retest outcome. Never include real passwords or payment credentials in evidence.